THE LONG EXACT COHOMOLOGY SEQUENCE OF THE RLIM TRIANGLE BREAKS INTO SHORT EXACT SEQUENCES

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Allowed standard prerequisites (editorial).

- The distinguished triangle computing Rlim K has cohomology long exact sequence with middle map the difference map between products of the H^p(K_n). invoked at statement lines 23303-23304 and proof lines 23315-23320. Provides the long exact sequence to be broken up.
- The kernel of the product-to-product map is lim H^p(K_n) and its cokernel is R^1 lim H^p(K_n), computed explicitly. invoked at proof lines 23321-23324. Identifies the kernel and cokernel of the middle map.

Source reference. Original lemma 07KX: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23270–23270. Cited at statement line 23303 for the notation and the triangle whose cohomology sequence is analyzed; the triangle is used inside the counted span.

Source reference. Original lemma 07KW: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22943–22943. Cited at proof lines 23322-23323 to identify the cokernel of the middle map; the conclusion is used inside the counted span.

Lemma 0.2. *With notation as in Lemma 07KX the long exact cohomology sequence associated to the distinguished triangle breaks up into short exact sequences*

$$0 \to R^1 \lim_n H^{p-1}(K_n^\bullet) \to H^p(R\lim K) \to \lim_n H^p(K_n^\bullet) \to 0$$

Proof. The long exact sequence of the distinguished triangle is

$$\ldots \to H^p(R\lim K) \to \prod\nolimits_n H^p(K_n^\bullet) \to \prod\nolimits_n H^p(K_n^\bullet) \to H^{p+1}(R\lim K) \to \ldots$$

The map in the middle has kernel $\lim_n H^p(K_n^\bullet)$ by its explicit description given in the lemma. The cokernel of this map is $R^1 \lim_n H^p(K_n^\bullet)$ by Lemma 07KW. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.